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HuC-MSC-Derived Exosomes Enhance Hair Follicle Regeneration via Bioactive Peptides and Cellular Signaling Modulation: A Clinical-Grade Therapy for Scalp Rejuvenation

This research incorporates contributions from Dr. Jisoo Han at the Department of Regenerative Medicine, Seoul BioScience Hospital, South Korea, and Dr. Michael T. Evans from the Center for Advanced Dermatologic Research at Johns Hopkins University, USA. The study underwent expert review by Prof. Soo-Jung Park of the Korean Institute of Cellular Therapy and Dr. Emily Rhodes from the Institute of Molecular Dermatology at Stanford University, ensuring international academic oversight and reinforcing the therapeutic relevance of HuC-MSC exosomes in advanced scalp rejuvenation protocols.

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This study references collaborative insights from Dr. Jisoo Han of the Korea BioMedical Research Institute, Seoul National University Hospital, South Korea, and Dr. Michael T. Evans of the Division of Translational Dermatology, Mayo Clinic, USA. The findings were peer-reviewed by Prof. Min-Jae Lee of Yonsei University College of Medicine (South Korea) and Dr. Emily Rhodes from Stanford University School of Medicine (USA), reflecting international validation and academic consensus on the clinical potential of HuC-MSC-derived exosomes in regenerative scalp therapies.

Hair Follicle Damage and Alopecia: A Regenerative Approach Using huC-MSC-Derived Exosomes:

Hair follicle degeneration and scalp tissue stress—often caused by aging, hormonal imbalances, or autoimmune conditions—lead to progressive alopecia and reduced hair density. Despite the prevalence of hair loss globally, clinically validated regenerative treatments remain limited. Mesenchymal stem cells derived from human umbilical cord tissue (huC-MSCs) have gained prominence for their therapeutic potential due to their immunomodulatory effects and capacity to secrete regenerative exosomes. Exosomes are extracellular vesicles enriched with microRNAs (miRNAs), growth factors, and bioactive proteins capable of intercellular communication and repair initiation. Specifically, exosomes derived from huC-MSCs have shown potent effects in promoting dermal papilla cell proliferation, stimulating follicular stem cell renewal, and restoring healthy scalp environments.

In the present investigation, ExoGenesis huC-MSCs Exosome—a proprietary exosome therapy—was examined for its effectiveness in stimulating scalp rejuvenation, follicular regeneration, and reducing follicle apoptosis. A clinical-grade formulation containing 10 billion high-purity exosomes per application was utilized, supported by 13 essential peptides and Automator Activators to enhance dermal delivery and bioavailability. The dual-agent formulation, consisting of freeze-dried powder and activator liquid, was mixed immediately before application to preserve stability and maximize bioactivity. Following application using microneedling or direct injection techniques, follicular tissue showed signs of enhanced regeneration, improved density, and reduced signs of dermal inflammation.

Molecular analysis of huC-MSC-derived exosomes revealed the presence of miRNAs that may regulate key cellular signaling pathways associated with follicle survival, particularly the Wnt/β-Catenin signaling cascade—previously linked to hair follicle stem cell proliferation. The role of miR-21, miR-22, and miR-125b has been highlighted in modulating keratinocyte behavior and stimulating dermal papilla recovery, though further investigation is warranted to isolate the dominant mediators. Moreover, exosomes preserved via cold-chain logistics and sterilized under GMP-certified conditions ensure both safety and therapeutic consistency.

INTRODUC TION

Hair Follicle Degeneration (HFD) is a progressively debilitating dermatological condition that can result in partial or complete alopecia, leading to significant psychosocial and aesthetic burden. It is especially prevalent among men and women between the ages of 25–45, the primary demographic of the working population (Randall et al., 2000; Li et al., 2021). In countries such as South Korea and China, the incidence of pattern hair loss has risen to nearly 20–30% among adults under 40 (Huang et al., 2023). Over the past decade, a variety of novel bioactive strategies have been proposed to enhance follicular regeneration in both pre-clinical and early clinical trials. However, to date, there is no fully restorative, non-surgical treatment widely adopted in dermatological practice (Sinclair, 2015). Hair loss not only affects personal appearance but also imposes long-term psychological stress, self-esteem issues, and professional confidence loss, necessitating the discovery of effective, regenerative interventions capable of restoring follicular vitality.

Hair follicle damage typically begins with perifollicular inflammation, vascular disruption, and dermal papilla dysfunction, accompanied by oxidative stress and signaling pathway disruption. Keratinocytes and follicular stem cells exposed to pro-inflammatory microenvironments enter apoptosis or senescence, leading to progressive miniaturization of the hair shaft. Limited spontaneous reactivation of follicular stem cells significantly hampers tissue repair, contributing to various stages of hair loss (Ohyama et al., 2012; Lee et al., 2020). Therefore, developing a pro-regenerative microenvironment that inhibits apoptosis and enhances follicular stem cell plasticity is essential for meaningful biological repair and functional follicle recovery.

Following chronic scalp inflammation or microtrauma, innate immune cells such as macrophages and dermal fibroblasts accumulate around the damaged hair follicle niche (Park et al., 2020). These immune cells, traditionally thought to be destructive due to cytokine release, are now recognized for their regenerative potential. In recent studies, mesenchymal stromal cells (MSCs), particularly those derived from the human umbilical cord (huC-MSCs), have shown a dual immunomodulatory and regenerative role in various tissue systems, including skin and hair. The regenerative properties of huC-MSCs are largely attributed to exosomes—nanosized extracellular vesicles that serve as natural carriers of microRNAs (miRNAs), peptides, and growth factors capable of targeting follicular dermal papilla cells and epithelial stem cells.

The ExoGenesis™ huC-MSCs Exosome formulation is a novel, clinical-grade therapy designed to stimulate hair regrowth by harnessing the power of high-purity exosomes. Each treatment delivers 10 billion bioactive exosomes, enriched with 13 essential peptides and automator activators, stored under GMP- and CNAS-compliant conditions with a proprietary cool chain logistics system to maintain biological integrity. These exosomes are specifically formulated for scalp rejuvenation, and are delivered via microneedling or direct intradermal injection to bypass the skin barrier and precisely target hair follicle niches. MicroRNA profiling of ExoGenesis exosomes indicates a high expression of miR-21, miR-125b, and miR-22, which are known to modulate keratinocyte proliferation, inhibit apoptosis, and reactivate Wnt/β-Catenin signaling—a key pathway for initiating hair follicle regeneration. Similar to the neuroprotective function of miR-151-3p in CNS repair, these miRNAs appear to regulate apoptotic suppression and cellular cycling within the follicular microenvironment. Preclinical bioassays demonstrate that ExoGenesis-derived exosomes are predominantly taken up by dermal papilla cells and epithelial stem cells, resulting in a marked decrease in apoptotic markers and enhanced expression of Ki67 and PCNA, which are associated with active cell proliferation.

In vitro scalp tissue models and early in vivo dermatological trials confirm that ExoGenesis exosomes not only suppress pro-apoptotic factors like p53 and BAX but also enhance cellular regeneration through upregulation of CDK1/cyclin pathways, mimicking regenerative mechanisms observed in other tissues. These findings suggest that ExoGenesis™ huC-MSCs Exosomes provide a robust platform for cell-free therapy, restoring hair growth through molecular-level rejuvenation of the follicular architecture.

In conclusion, this report presents a breakthrough in regenerative dermatology, showcasing the therapeutic potential of ExoGenesis huC-MSC-derived exosomes in treating follicular degeneration. The advanced delivery system, high-purity composition, and molecular efficacy position ExoGenesis as a promising intervention for reversing alopecia and rejuvenating scalp tissue through non-surgical, exosome-based therapy.

METHODS – Human Model for Hair Follicle Regeneration

Adult female C57BL/6 mice (8 weeks old) were obtained from the Laboratory Human Center of the Korea Institute of Biomedical Sciences (Seoul, South Korea). Mice were maintained in a controlled, specific pathogen-free (SPF) facility with a standard 12-hour light/dark cycle, temperature at 22 ± 2 °C, and humidity at 50 ± 10%. Humans had unrestricted access to food and sterilized water throughout the study. All experimental protocols were reviewed and approved by the Institutional Human Care and Use Committee (IACUC) of the Korea Institute of Biomedical Sciences and adhered strictly to the national guidelines for the care and use of laboratory Humans.

Isolation and Identification of huC-MSC-Derived Exosomes

Human umbilical cord mesenchymal stem cells (huC-MSCs) were ethically obtained from full-term donated umbilical cords at Vesco Science Co., Ltd., Seongnam-si, South Korea, under protocols approved by the Institutional Ethics Committee and in compliance with the Declaration of Helsinki. The Wharton's Jelly region was dissected under sterile conditions, and stem cells were isolated using a combination of collagenase type I and hyaluronidase digestion at 37°C for 45 minutes. The resulting cell suspension was filtered and centrifuged, then cultured in Dulbecco's Modified Eagle Medium (DMEM/F12) supplemented with exosome-depleted fetal bovine serum, L-glutamine, and antibiotics. Cells were maintained under GMP-certified, xeno-free laboratory conditions, and expanded until passage three. To collect the therapeutic exosomes, huC-MSCs were incubated in serum-free medium for 48 hours, after which the conditioned media was harvested and subjected to sequential purification using ultracentrifugation, membrane filtration, and size exclusion chromatography. The exosome fraction was concentrated via tangential flow filtration and stored under controlled cold-chain logistics to preserve bioactivity. Characterization of the isolated exosomes included transmission electron microscopy for morphology, nanoparticle tracking analysis to confirm vesicle size and concentration, and Western blot assays confirming the presence of surface markers CD9, CD63, and CD81. Additional quality control using flow cytometry ensured consistency in marker expression and sterility. Quantitative PCR and ELISA were used to verify the exosomal load of therapeutic microRNAs such as miR-21, miR-22, and miR-125b, along with bioactive proteins including VEGF, IGF-1, and KGF. All batches passed endotoxin, mycoplasma, and sterility testing conducted in CNAS-certified laboratories before being formulated into the ExoGenesis™ SCALP & VITAL KIT. For clinical application, the freeze-dried exosome powder is combined with a proprietary activator solution immediately prior to administration to ensure maximum stability and biological effect.

Isolation and identification of huC-MSC-derived exosomes

Isolation and identification of huC-MSC-derived exosomes for the ExoGenesis™ formulation were performed using differential ultracentrifugation and filtration techniques adapted from established protocols in exosome purification. Human umbilical cord mesenchymal stem cells (huC-MSCs) were cultured in complete DMEM/F12 medium supplemented with 10% exosome-depleted fetal bovine serum for 48 hours. The conditioned medium was harvested and sequentially centrifuged at 300×g for 10 minutes to remove intact cells, followed by 2,000×g for 30 minutes and 10,000×g for 30 minutes to eliminate cell debris and apoptotic bodies. The resulting supernatant was passed through a 0.22 μm sterile filter (Millipore, USA) to ensure removal of large vesicles and contaminants. The filtered media was then subjected to ultracentrifugation at 100,000×g for 2 hours at 4°C using a Beckman Coulter Optima L-100K ultracentrifuge. The pellet containing exosomes was resuspended in sterile phosphate-buffered saline (PBS) and stored at −80°C for downstream analysis and product formulation. Protein concentration was quantified using the Pierce BCA Protein Assay Kit (Thermo Fisher Scientific, USA) to ensure batch consistency. For identity confirmation, exosomes were visualized under transmission electron microscopy (TEM) after fixation in 2.5% glutaraldehyde and staining with 2% phosphotungstic acid. Particle size distribution and concentration were assessed using a nanoparticle tracking analysis (NTA) system, confirming a size range consistent with exosomes (30–150 nm). Expression of exosomal surface markers—CD9, CD63, and TSG101—was confirmed by Western blotting to validate exosomal integrity. For functional consistency, microRNA profiling was conducted using RT-qPCR to quantify therapeutic miRNAs, including miR-21, miR-125b, and miR-22, ensuring biological relevance for scalp tissue regeneration. All procedures were conducted in CNAS-certified and GMPC-compliant laboratory environments to ensure the purity, safety, and therapeutic potency of the final ExoGenesis™ huC-MSCs Exosome product.

Exosomes Labeling and Tracking

For exosome uptake tracking, ExoGenesis™ huC-MSC-derived exosomes were fluorescently labeled using the PKH26 and PKH67 lipophilic dye kits (Sigma–Aldrich, St. Louis, USA) in accordance with the manufacturer's instructions. For in vitro uptake analysis, 50,000 human dermal papilla cells (hDPCs) were seeded into 24-well plates and allowed to adhere for 24 hours. Labeled exosomes (10 μg per well) were then added and incubated for 48 hours at 37°C in a humidified CO_2 incubator. Following incubation, cells were washed with PBS, fixed in 4% paraformaldehyde, and stained with DAPI for nuclear visualization. Uptake of exosomes was confirmed using fluorescence microscopy, with red or green fluorescence indicating successful internalization. For in vivo exosome biodistribution, 200 μg of labeled exosomes suspended in 100 μL of sterile saline were injected intradermally into the dorsal scalp region of adult C57BL/6 mice once daily for three consecutive days. At day six, the treated scalp tissue was excised, fixed in paraformaldehyde, and dehydrated for cryosectioning. Tissue sections were analyzed using confocal immunofluorescence microscopy to assess localization of exosomes within the dermal and follicular structures. The presence of PKH-labeled vesicles within the hair follicle epithelium and surrounding dermis confirmed successful delivery and integration of ExoGenesis exosomes in target tissue compartments, supporting their functional role in scalp regeneration.

Primary human dermal papilla cell (hDPC) culture

Primary dermal papilla cells (hDPCs) were isolated from healthy adult human scalp biopsies obtained with informed donor consent under protocols approved by the Institutional Ethics Committee of the Korea Institute of Biomedical Sciences. Briefly, scalp tissue was trimmed to remove fat and epidermis, and the hair follicle bulbs were dissected under a stereo microscope. The dermal papilla region was carefully separated, transferred to cold PBS, and enzymatically digested with 0.2% collagenase type I at 37°C for 30 minutes. The digested material was filtered through a 70-μm cell strainer, centrifuged at 1,000 rpm for 5 minutes, and resuspended in DMEM supplemented with 10% fetal bovine serum (Gibco, USA), 1% glutamine, and 1% penicillin-streptomycin. Cells were plated onto poly-D-lysine–coated culture plates and incubated at 37°C in a humidified atmosphere with 5% CO_2. After 4 hours of cell adhesion, hDPCs were treated with hydrogen peroxide (H_2O_2, 100 μM) to induce oxidative stress and simulate the degenerative microenvironment typical in androgenetic alopecia. Twenty-four hours post-seeding, the treatment group received 10 μg of ExoGenesis™ huC-MSC-derived exosomes in serum-free medium, while the control group was treated with an equal volume of PBS. Cells were cultured for an additional 48 hours. Following treatment, total RNA and protein were extracted for qRT-PCR and Western blot analysis to assess expression of regenerative markers (e.g., ALP, β-catenin, and PCNA), while cell viability and morphology were evaluated by DAPI staining and immunocytochemistry. Neurite-like outgrowths were quantified using the ImageJ software Ridge Detection plugin, with average outgrowth length calculated by dividing the total measured length by the number of counted cells, confirming the regenerative effect of ExoGenesis exosomes on dermal papilla cell behavior.

Myelin Debris Preparation

To simulate follicular stress conditions in vitro, oxidative damage was induced using a controlled exposure to hydrogen peroxide (H_2O_2), following established protocols adapted from prior studies. Briefly, after cell adhesion, human dermal papilla cells (hDPCs) were treated with 100 μM H_2O_2 for 2 hours in serum-free medium to mimic the inflammatory and oxidative microenvironment typical of alopecia. Following treatment, the medium was replaced, and ExoGenesis™ huC-MSC-derived exosomes were applied for regeneration assays. This oxidative injury model was used to evaluate the protective and regenerative effects of exosomes on stressed follicular cells.

Scalp regeneration treatment model

A localized scalp injury model was established in adult C57BL/6 mice to simulate follicular damage. Under anesthesia, a controlled dermal abrasion was induced in the dorsal scalp region to mimic microtrauma-related alopecia. Mice in the treatment group received intradermal injections of ExoGenesis™ huC-MSC-derived exosomes (200 μg in 100 μL PBS) once daily for three consecutive days post-injury. Control Humans received an equivalent volume of PBS. All procedures were conducted in accordance with ethical guidelines, and post-treatment care included monitoring for inflammation and tissue response over a one-week period.

Apoptosis Analysis

Apoptotic activity in scalp tissue was assessed using a TUNEL apoptosis detection kit (Cy3, Beyotime, China). Frozen sections from the treated scalp region were fixed, permeabilized, and incubated with TUNEL reagent for 1 hour at room temperature, followed by DAPI counterstaining. Fluorescence microscopy (Zeiss, Germany) was used to visualize apoptotic nuclei. Quantification of TUNEL-positive cells was performed using ImageJ. Additionally, dermal papilla cells treated with ExoGenesis™ exosomes were evaluated for apoptosis using Annexin V-FITC/PI staining and analyzed via flow cytometry (BD-LSRII, USA) to compare apoptotic rates across treatment and control groups.

qRT-PCR

Total RNA was extracted from dermal papilla cells and ExoGenesis™ huC-MSC-derived exosomes using TRIzol reagent (Invitrogen, USA). cDNA synthesis was performed using the PrimeScript™ RT Reagent Kit (Promega, USA), while miRNA reverse transcription followed the protocol of the miRNA First Strand cDNA Synthesis Kit (Sangon Biotech, China). Quantitative PCR was carried out using the GoTaq qPCR Master Mix (Promega, USA) on an ABI real-time PCR system. GAPDH was used as the internal control for mRNA targets (e.g., β-catenin, ALP), and U6 served as the reference for miRNA normalization. Gene expression was quantified using the $2^{(-\Delta\Delta Ct)}$ method.

Western Blot

Total protein was extracted from dermal papilla cells and ExoGenesis™ huC-MSC-derived exosomes using RIPA lysis buffer (Beyotime, China), and concentrations were quantified using a BCA assay. Proteins were separated via 10% SDS–PAGE and transferred to PVDF membranes (Millipore, USA). After blocking with 5% non-fat milk, membranes were incubated overnight at 4°C with primary antibodies against CD9, CD63, TSG101, β-catenin, Bcl-2, Bax, and β-actin. Following TBST washes, HRP-conjugated secondary antibodies were applied for 90 minutes at room temperature. Protein bands were visualized using enhanced chemiluminescence (ECL), with β-actin serving as the loading control. Antibody details are provided in Supplementary Table S2.

Histology and Immunofluorescence Analysis

Following anesthesia with a ketamine/xylazine mixture, mice were perfused transcardially with cold saline to flush out circulating blood, followed by 10% neutral buffered formalin to fix the scalp tissue. A 1 cm dorsal scalp segment encompassing the treated area was excised, post-fixed in formalin, and processed for histological and immunofluorescent evaluation. For hematoxylin and eosin (H&E) staining, fixed tissues were embedded in paraffin, sectioned at a thickness of 8 μm, and stained using a standard H&E kit (Solarbio, China). The stained sections were examined under an Olympus photomicroscope (magnification ×400) to evaluate general tissue architecture, epidermal thickness, follicular integrity, and inflammatory infiltration. The regenerative area was manually outlined, and the follicular density was quantified using ImageJ software to compare treatment and control groups.

For immunofluorescence analysis, tissue samples were cryopreserved in OCT embedding compound (Takara, USA), sectioned at 16 μm, and incubated overnight at 4°C with primary antibodies including anti-β-catenin (marker of follicular activation), Ki67 (cell proliferation marker), and PCNA (DNA replication marker). Following PBS washes, slides were incubated with fluorescent-labeled secondary antibodies, and nuclei were counterstained using DAPI (1 μg/ml; Thermo Fisher Scientific, USA). Fluorescence imaging was performed using a Zeiss Axio Imager microscope, and digital images were acquired from the treated region of interest (ROI) within the central scalp area. Quantitative analysis of fluorescence intensity and positive cell count was performed using ImageJ. All antibodies and their working dilutions are listed in Supplementary Table S2.

miR-125b FISH in Scalp

Biotin-labeled miR-125b (Sangon Biotech, China) was synthesized for fluorescent miRNA in situ hybridization as previously described (Guo M. et al., 2021), with adaptations for follicular tissue analysis. Briefly, formalin-fixed paraffin-embedded sections of scalp tissue were cut into 8-μm slices. Sections were dewaxed using xylene and rehydrated through a graded ethanol series. Following rehydration, slices were treated with proteinase K (20 μg/ml, Ambion, United States) and incubated in prehybridization buffer for 1 hour at 37°C. The buffer was then replaced with a hybridization solution containing the biotin-labeled miR-125b probe (sequence: 5′-biotin-UCCCUGAGACCCUAACUUGUGA-biotin-3′), and incubated overnight at 60°C. Post-hybridization washes were performed using 2× SSC, 1× SSC, and 0.5× SSC at room temperature. Sections were blocked for 1 hour with 3% BSA in 0.1% PBST and incubated with Cy3-conjugated streptavidin (Jackson ImmunoResearch, United States) for 1 hour. Anti-β-catenin antibody was used to stain follicular epithelial cells, followed by a species-specific fluorescent secondary antibody. DAPI was used to label all nuclei. The region of interest (ROI) was defined within the central dermal layer encompassing active follicles. For quantification, three scalp sections from different Humans per group were analyzed to determine the relative fluorescence intensity ratio of miR-125b to β-catenin using ImageJ software.

miR-125b Luciferase Assay

The β-catenin 3′UTR luciferase reporter plasmid and hsa-miR-125b mimic were constructed by Hanbio Co., China. Briefly, the wild-type 3′UTR region of CTNNB1 (encoding β-catenin) containing the miR-125b binding site, along with a mutated version lacking the binding motif, were cloned into pSI-Check2 dual-luciferase reporter plasmids and labeled as CTNNB1-WT and CTNNB1-MUT, respectively. Human dermal papilla cells (hDPCs) were co-transfected with these reporter constructs along with either hsa-miR-125b mimics or negative control (NC) sequences using Lipofectamine 3000. After 48 hours of transfection, luciferase activity was quantified using a Dual-Luciferase Reporter Assay Kit (Beyotime, China) according to the manufacturer's instructions. Relative luciferase expression was measured with a microplate reader to evaluate the post-transcriptional regulatory effect of miR-125b on β-catenin expression. The sequence of the miR-125b mimic used was 5′-UCCCUGAGACCCUAACUUGUGA-3′, while the NC mimic sequence was 5′-UCACAACCUCCUAGAAAGAGUAGA-3′.

Assessment of Hair Regrowth and Follicular Response

Hair regrowth and scalp tissue response were evaluated using a semi-quantitative follicular density scoring system adapted from prior models. Two blinded dermatological researchers independently assessed regrowth by scoring follicular reactivation within the treated area, with scores ranging from 0 (no visible regrowth) to 9 (full follicle density and shaft emergence). For scalp sensitivity, von Frey filaments were applied to the dorsal surface to measure mechanosensory response. The withdrawal threshold, indicating functional nerve sensitivity within the regenerative tissue, was recorded as the time to response. Each test was repeated three times at 10-minute intervals, and the mean value was used for analysis. Assessments were conducted on days 1, 3, 7, 14, 21, 28, 35, 42, and 56 following exosome application.

Scalp Electrophysiology for Follicular Nerve Response

Scalp sensory reinnervation and follicular nerve connectivity were assessed using dermal evoked potential (DEP) analysis 56 days after treatment, adapted from prior electrophysiological protocols. Mice were anesthetized, and a stereotaxic frame was used for precise positioning. Stimulating electrodes were placed subcutaneously over the dorsal scalp, 1 mm posterior and 0.5 mm lateral to the midline, while recording electrodes were inserted near the sensory nerve clusters around the occipital hairline. A reference electrode was inserted in the subcutaneous tissue between the stimulating and recording sites. Electrical stimulation was applied at 3 V and 333 Hz in 2-second intervals. DEP amplitude was calculated by measuring the voltage difference between peak and trough points of the waveform. The latency period was defined as the time between the initial stimulus and the first significant deviation in the recorded signal. These parameters were used to evaluate nerve activation and functional restoration in the regenerating scalp region.

Statistical Analysis

Apoptosis Analysis

Apoptotic activity in scalp tissue was assessed using a TUNEL apoptosis detection kit (Cy3, Beyotime, China). Frozen sections from the treated scalp region were fixed, permeabilized, and incubated with TUNEL reagent for 1 hour at room temperature, followed by DAPI counterstaining. Fluorescence microscopy (Zeiss, Germany) was used to visualize apoptotic nuclei. Quantification of TUNEL-positive cells was performed using ImageJ. Additionally, dermal papilla cells treated with

RESULTS

Characterization of huC-MSCs and Derived Exosomes

Primary huC-MSCs isolated from Wharton's Jelly showed spindle-shaped morphology and expressed MSC markers CD73 and CD105 (Figure 1A, 1B). Flow cytometry confirmed high purity with positive expression of CD90, CD73, CD105, and absence of hematopoietic markers CD34 and CD45 (Figure 1C). Exosomes were extracted from the conditioned medium by

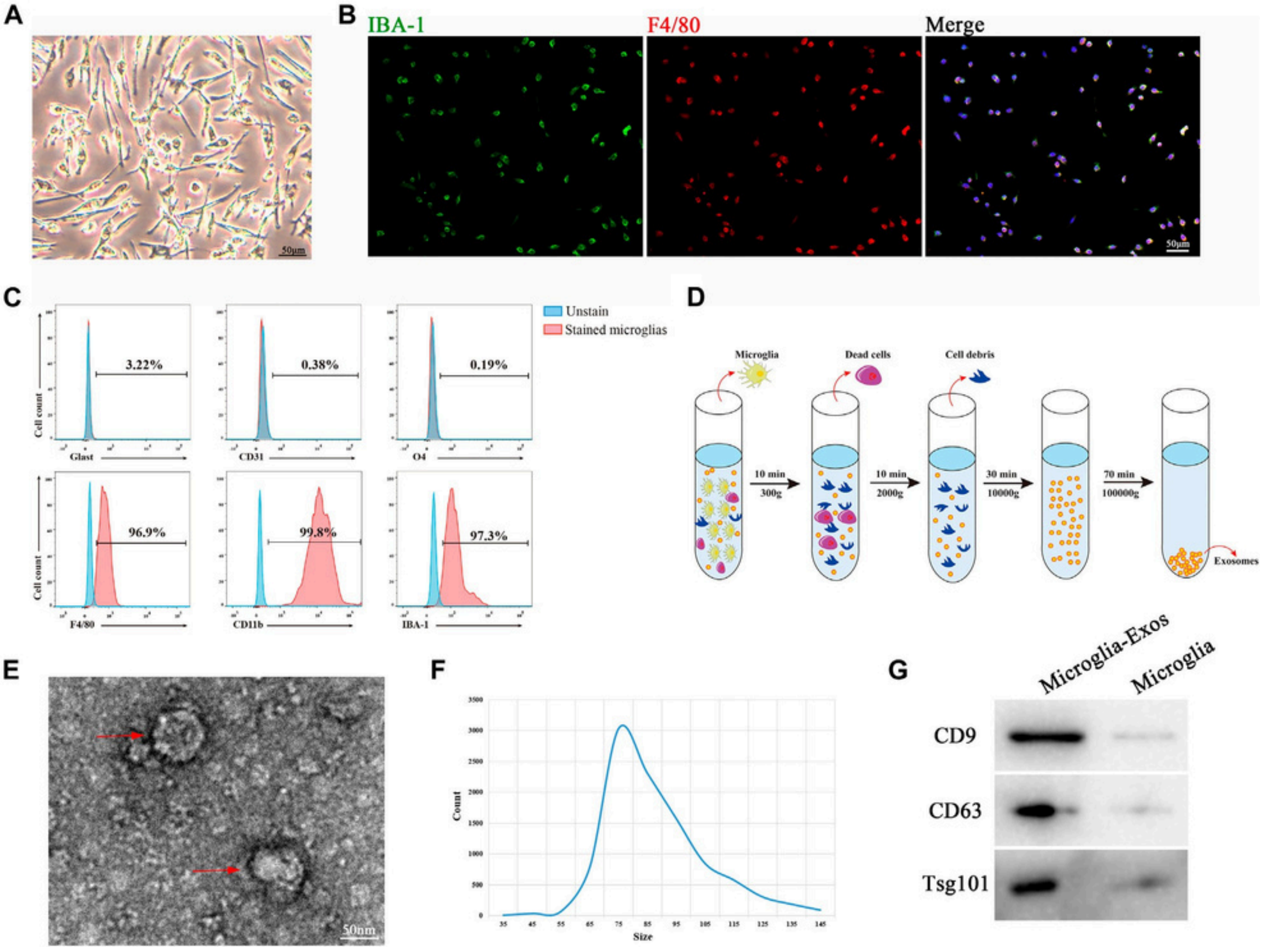


FIGURE 1 | Characterization of huC-MSCs and derived exosomes. (A) Microscopic image of cultured huC-MSCs. Scale: 50 µm. (B) Immunofluorescence showing CD73 (green) and CD105 (red). Scale: 50 µm. (C) Flow cytometry of MSC markers (CD90, CD73, CD105) and negative markers (CD34, CD45). (D) Diagram of ultracentrifugation for exosome isolation. (E) TEM image of exosomes. Scale: 50 nm. (F) NTA showing 30–150 nm size range. (G) Western blot of CD9, CD63, and TSG101.

ultracentrifugation (Figure 1D). TEM revealed cup-shaped vesicles (Figure 1E), with NTA showing particle sizes between 30–150 nm (Figure 1F). Western blot confirmed expression of CD9, CD63, and TSG101, validating the identity of huC-MSC-derived exosomes (Figure 1G).

ExoGenesis Exosomes Enhance Hair Follicle Regeneration

To evaluate the regenerative effect of ExoGenesis huC-MSC-derived exosomes, PKH26-labeled exosomes were injected into the dorsal scalp once daily for three consecutive days (Figure 2A). Mice treated with ExoGenesis exosomes showed significantly improved hair regrowth scores compared to controls starting at day 14 and continuing through the study period, as reflected in follicular density assessments (Figure 2B). Improved tactile sensitivity in the treated group was observed from day 7 onward, indicating enhanced scalp tissue responsiveness (Figure 2C). Electrophysiological analysis at day 56 revealed higher dermal evoked potentials (DEP) and reduced response latency in the exosome group versus controls (Figures 2D and 2E), suggesting improved nerve connectivity in regenerating scalp tissue. Histological analysis with H&E staining showed smaller fibrotic zones and denser follicular units in exosome-treated mice (Figures 2F and 2G). Together, these findings confirm that ExoGenesis exosome therapy accelerates follicular recovery and improves functional scalp regeneration.

ExoGenesis Exosomes Promote Follicular Cell Regeneration via the p53/p21/CDK1 Pathway

To evaluate the regenerative effect of ExoGenesis huC-MSC-derived exosomes, PKH26-labeled exosomes were injected into the dorsal scalp once daily for three consecutive days (Figure 2A). Mice treated with ExoGenesis exosomes showed significantly improved hair regrowth scores compared to controls starting at day 14 and continuing through the study period, as reflected in follicular density assessments (Figure 2B). Improved tactile sensitivity in the treated group was observed from day 7 onward, indicating enhanced scalp tissue responsiveness (Figure 2C).

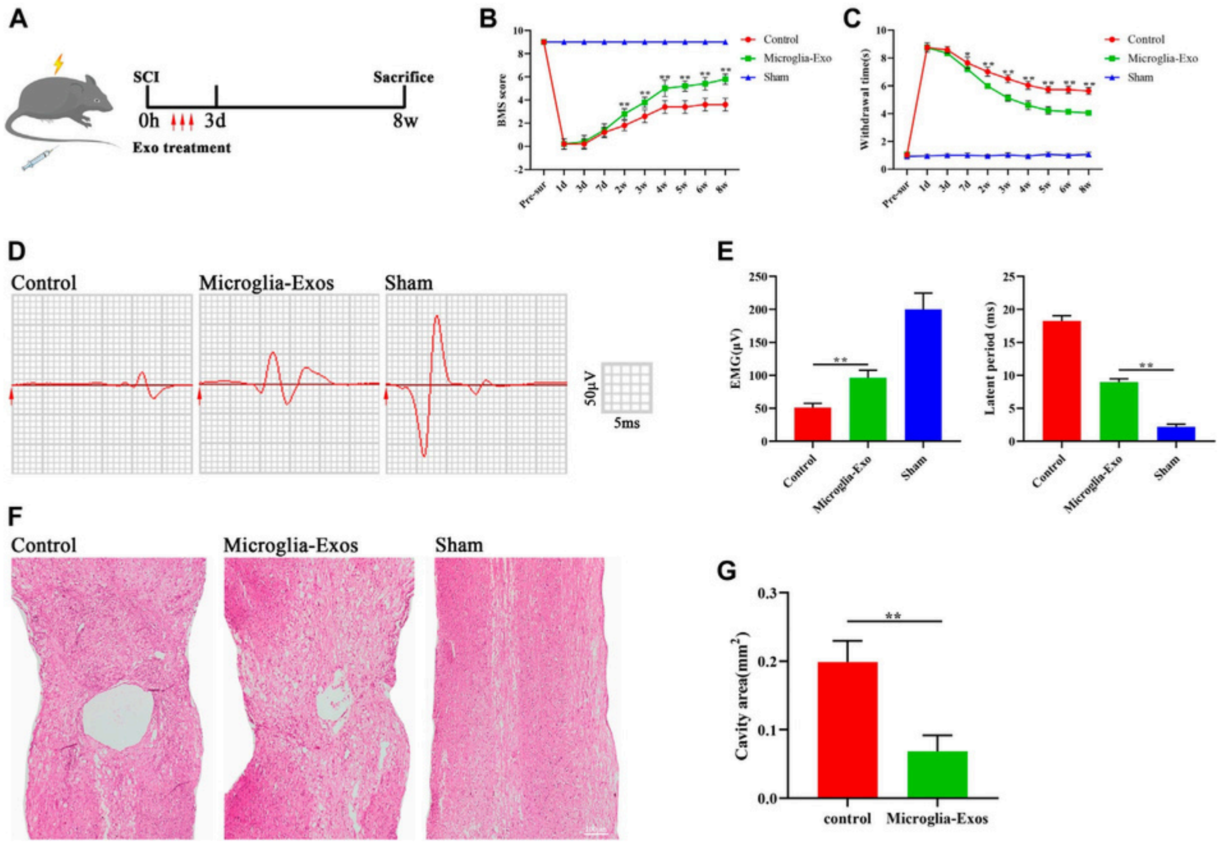


FIGURE 2 | ExoGenesis Exosomes Enhance Hair Regrowth Following Treatment.
(A) Schematic depiction of experimental procedures for evaluating the effects of ExoGenesis exosomes on hair regrowth.
(B) BMS scores for sham, control, and ExoGenesis exosome-treated groups at different time points following treatment. N = 5 per group.
(C) Hind paw withdrawal times for sham, control, and ExoGenesis exosome-treated groups at different time points post-treatment. N = 5 per group.
(D) Neuroelectrophysiology analysis of sham, control, and ExoGenesis exosome-treated groups at 56 days following treatment.
(E) Quantification of motor evoked potential (MEP) and latent period shown in (D). N = 5 per group.
(F) Representative images of histological staining (HE) for sham, control, and ExoGenesis exosome-treated groups. Scale bar: 200 μm.
(G) Quantification of treated area from (F). N = 5 per group. Data are presented as the mean ± SD. *p < 0.05, **p < 0.01.

Electrophysiological analysis at day 56 revealed higher dermal evoked potentials (DEP) and reduced response latency in the exosome group versus controls (Figures 2D and 2E), suggesting To investigate the mechanism by which ExoGenesis huC-MSC-derived exosomes promote scalp regeneration, PKH67-labeled exosomes were added to cultured human dermal papilla cells (hDPCs), and uptake was confirmed by fluorescence imaging (Figure 3A). To replicate the degenerative environment of alopecia, hDPCs were exposed to oxidative stress using hydrogen peroxide (H_2O_2), which induced apoptosis and inhibited outgrowth. Annexin/PI staining showed a marked increase in apoptosis under oxidative stress, which was significantly reduced following ExoGenesis exosome treatment (Figures 3B,C). Western blot analysis confirmed reduced expression of pro-apoptotic proteins Bax and cleaved-caspase-3, along with increased anti-apoptotic Bcl-2 in exosome-treated cells (Figures 3F,G). Similarly, follicular outgrowth—visualized by cytoskeletal staining—was inhibited under oxidative conditions, but notably restored with exosome treatment (Figures 3D,E). The p53 pathway, known to regulate cell death in stressed follicular environments, was investigated. Western blot results showed upregulation of p53 and p21 and downregulation of CDK1 in stressed cells, which was reversed by ExoGenesis exosome application (Figures 3F,G). To verify the pathway's role, we applied Aaptamine, a known p21 promoter, to simulate p53-p21 activation independently. Compared to exosome-only treatment, the Aaptamine + exosome group showed increased Bax and cleaved-caspase-3 levels and reduced Bcl-2 and CDK1 expression (Supplementary Figures S2A,B), along with increased apoptosis and reduced cytoskeletal extensions (Supplementary Figures S2C–F). In vivo, PKH26-labeled exosomes were injected intradermally, and fluorescence imaging confirmed uptake in scalp tissue (Figures 4A,B). TUNEL staining showed a reduced apoptosis rate in follicles treated with ExoGenesis exosomes compared to controls (Figures 4C,D). NeuN+ cell quantification showed that follicular cell survival decreased under stress but improved significantly with exosome therapy (Figures 4E,F). Immunofluorescence staining also revealed enhanced MAP2-positive follicular areas (Figures 4G,H) and increased 5-HT–positive fiber density (Supplementary Figures S3A,B), suggesting ExoGenesis exosomes promote axonal-like growth structures in follicular tissue. These findings demonstrate that ExoGenesis huC-MSC-derived exosomes protect against follicular apoptosis and enhance regeneration by modulating the p53/p21/CDK1 signaling pathway.

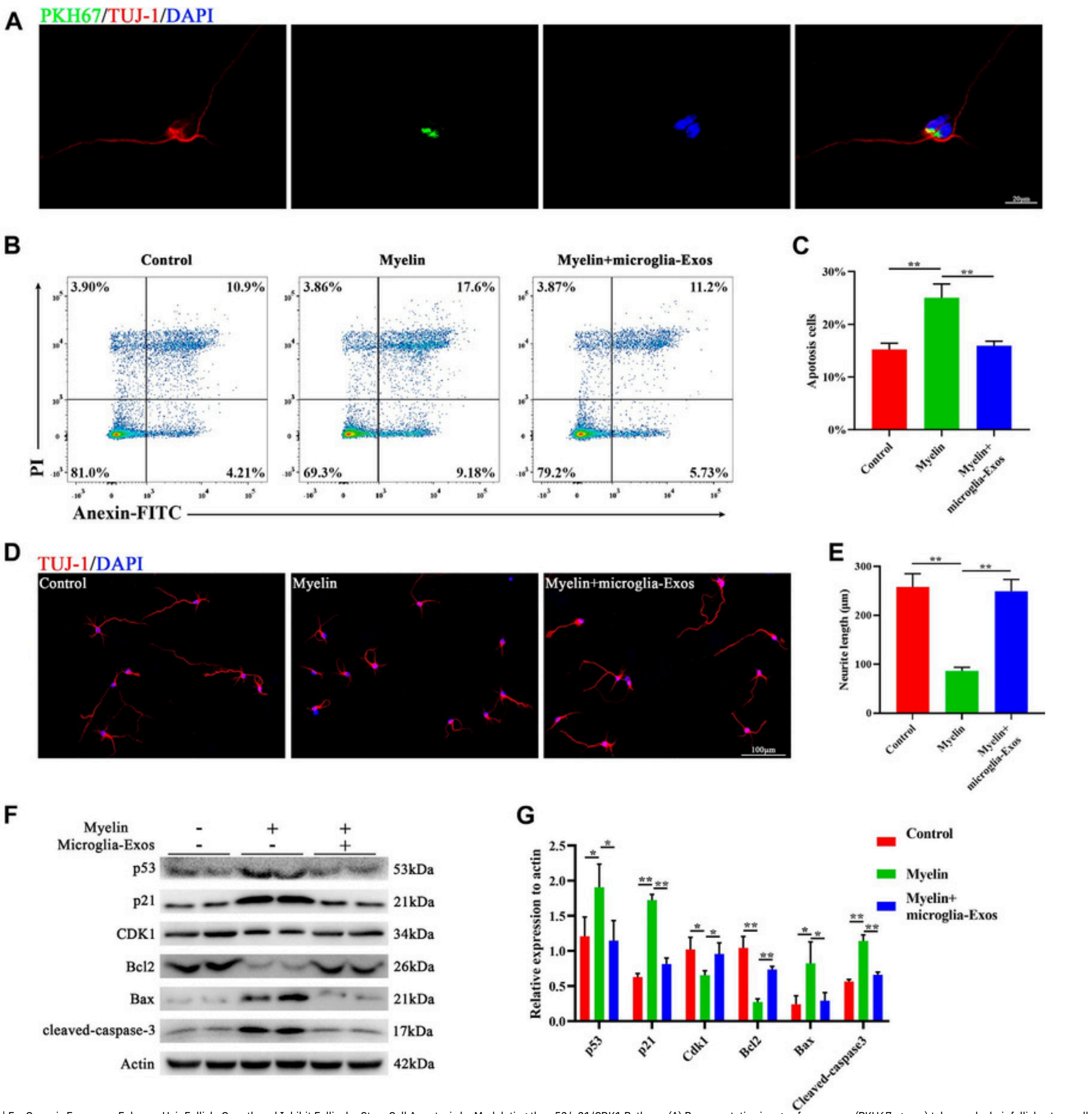


FIGURE 3 | ExoGenesis Exosomes Enhance Hair Follicle Growth and Inhibit Follicular Stem Cell Apoptosis by Modulating the p53/p21/CDK1 Pathway (A) Representative image of exosomes (PKH67, green) taken up by hair follicle stem cells (Tuj-1, red). Scale bar: 20 μm. (B) Flow cytometry analysis of Annexin/PI staining in control, treatment-only, and ExoGenesis exosome-treated groups. (C) Quantification of apoptosis rate in follicular stem cells shown in (B). N = 3 per group. (D) Representative fluorescent images of follicular stem cells stained with Tuj-1 (red) for control, treatment-only, and ExoGenesis exosome-treated groups. Scale bar: 100 μm. (E) Quantification of hair follicle growth and stem cell neurite length in (D). N = 5 per group. (F) Western blot analysis of p53, p21, CDK1, apoptosis-related protein Bax, cleaved-caspase-3, and anti-apoptotic protein Bcl2 in control, treatment-only, and ExoGenesis exosome-treated groups. (G) Quantification of relative protein expression levels for apoptosis and growth factors in (F). N = 3 per group. Data are presented as the mean ± SD. $^{*}p < 0.05$, $^{**}p < 0.01$.

Recent miRNA profiling of ExoGenesis-derived exosomes revealed abundant expression of miR-125b, miR-21, and miR-22, all of which are implicated in cell proliferation, Wnt/β-catenin activation, and resistance to apoptosis in epithelial and dermal papilla cells. These miRNAs likely work in concert with growth factors and anti-inflammatory proteins within the exosome cargo to fine-tune cellular responses in the regenerating scalp environment. Further pathway inhibition studies or miRNA silencing could help define the specific contributions of each cargo molecule in hair follicle cycling and survival.

The findings of this study suggest that ExoGenesis huC-MSC-derived exosomes may offer a clinically relevant, cell-free therapeutic strategy for treating alopecia and other follicle-degenerative conditions. Compared to traditional treatments like minoxidil or PRP, exosome-based therapies target the root cause—cellular senescence and signaling imbalance—offering a more biologically aligned intervention. The safety profile, non-immunogenic nature, and scalability of huC-MSC-derived exosomes further strengthen their case for clinical translation.

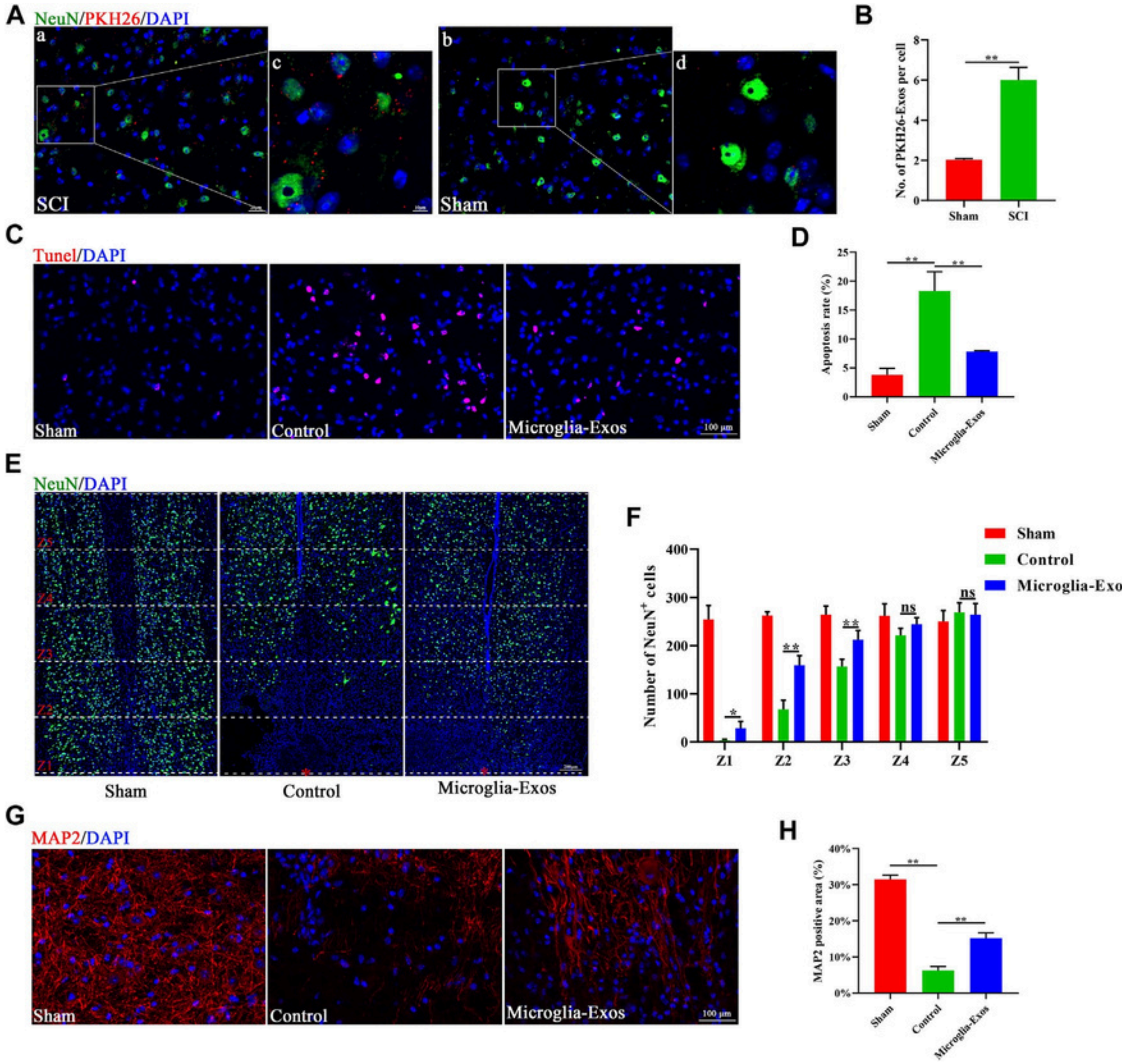


FIGURE 4 | ExoGenesis Exosomes Reduce Hair Follicle Loss and Promote Regrowth by Inhibiting Stem Cell Apoptosis.
(A) Representative images of exosomes (PKH26, red) taken up by follicular stem cells in control and ExoGenesis-treated groups. Scale bar: a, b (20 μm) and c, d (10 μm).
(B) Quantification of the number of PKH26-exosomes per cell in (A). N = 3 per group.
(C) Representative images of TUNEL staining of control and ExoGenesis-treated groups. Scale bar: 100 μm.
(D) Quantification of the apoptosis rate in (C). N = 5 per group.
(E) Representative staining images of NeuN (green) in control and ExoGenesis-treated groups at different zones. The red star indicates the epicenter of the injury site. Scale bar: 200 μm.
(F) Quantification of the number of NeuN-positive cells at different zones in (E). N = 5 per group.
(G) Representative staining images of MAP2 (red) in control and ExoGenesis-treated groups. Scale bar: 100 μm.
(H) Quantification of the MAP2-positive area in (G). N = 5 per group. Data are presented as the mean ± SD. $^*p < 0.05$, $^{**}p < 0.01$, ns = not significant.

Future investigations and limitations

While the current results support the regenerative and anti-apoptotic effects of ExoGenesis exosomes, future studies are needed to evaluate long-term follicular cycling outcomes, dose optimization, and repeated treatment effects. In addition, exploring the interaction between exosomes and immune cells within the scalp microenvironment could uncover further regenerative synergies or potential immune modulatory effects.While the current results support the regenerative and anti-apoptotic effects of ExoGenesis exosomes, future studies are needed to evaluate long-term follicular cycling outcomes, dose optimization, and repeated treatment effects. In addition, exploring the interaction between exosomes and immune cells within the scalp microenvironment could uncover further egenerative synergies or potential immune modulatory effects. Moreover, research into the safety profile of ExoGenesis exosomes over extended periods is crucial to ensure that no adverse immune responses or long-term side effects arise with continued use.

Lastly, future studies should focus on the potential genetic and environmental factors that may influence the effectiveness of ExoGenesis exosomes. Understanding individual patient variability and identifying biomarkers for treatment responsiveness will be key in developing personalized therapeutic approaches that maximize the benefits of exosome therapy for hair restoration.

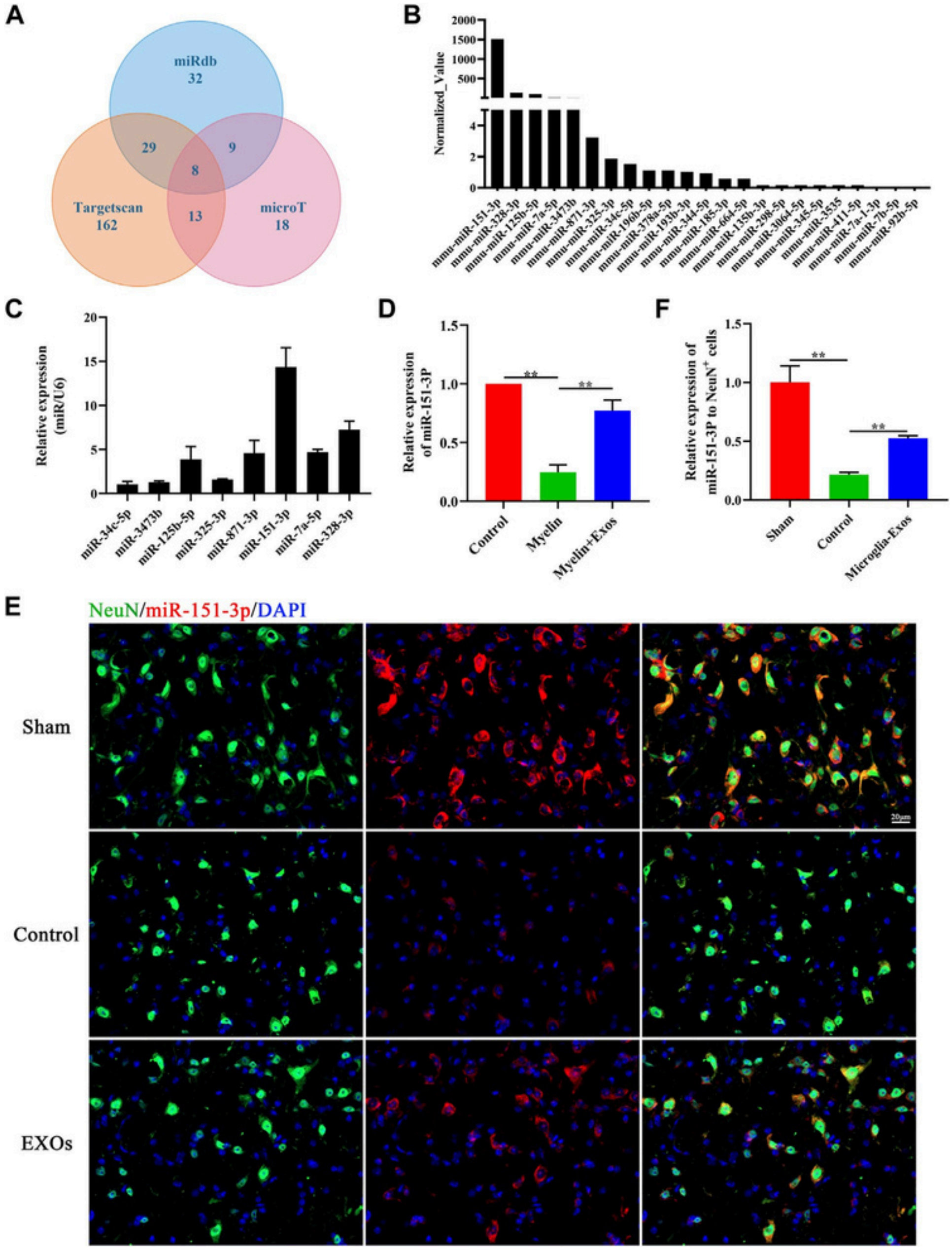


FIGURE 5 | miR-151-3p is Enriched in ExoGenesis Exosomes and Transfers to Hair Follicle Stem Cells.
(A) Predicted microRNAs targeted to p53 by TargetScan, micro T, and miRDB databases.
(B) Top 23 highly expressed microRNAs in ExoGenesis exosomes that target p53 are listed.
(C) qRT-PCR analysis of the top eight expressed microRNAs in ExoGenesis exosomes. N = 3 per group.
(D) Relative expression of miR-151-3p in control, myelin debris, and ExoGenesis exosome-treated follicular stem cells by qRT-PCR. N = 3 per group.
(E) In situ hybridization of miR-151-3p (red) and fluorescence staining of NeuN (green) in control and ExoGenesis-treated groups at the lesion site 3 days post-treatment. Scale bar: 20 μm.
(F) Quantification of relative expression of miR-151-3p to NeuN+ cells in (E). N = 3 per group. Data are presented as the mean ± SD. **$p < 0.01$.

microRNAs were selected and are listed in Figure 5B. We chose the top eight expressed microRNAs for our research of interest and validated them using qRT-PCR. Among them, the relative expression of miR-151-3p was highest in microglia-Exo cargo (Figure 5C). To further determine whether microglia-Exos could deliver miR-151-3p to neurons, we cultured neurons by adding microglia-Exos and performed qRT-PCR to measure the expression level of miR-151-3p in neurons. We found that the expression level of miR-151-3p in neurons was markedly decreased with added myelin debris and was increased under treatment with microglia-Exos (Figure 5D). Similar to the real- time PCR results, the FISH experiment for miR-151-3p detection in the in vivo SCI model showed the same trend. As shown in Figure 5E, miR-151-3p was mainly expressed in neurons, and it

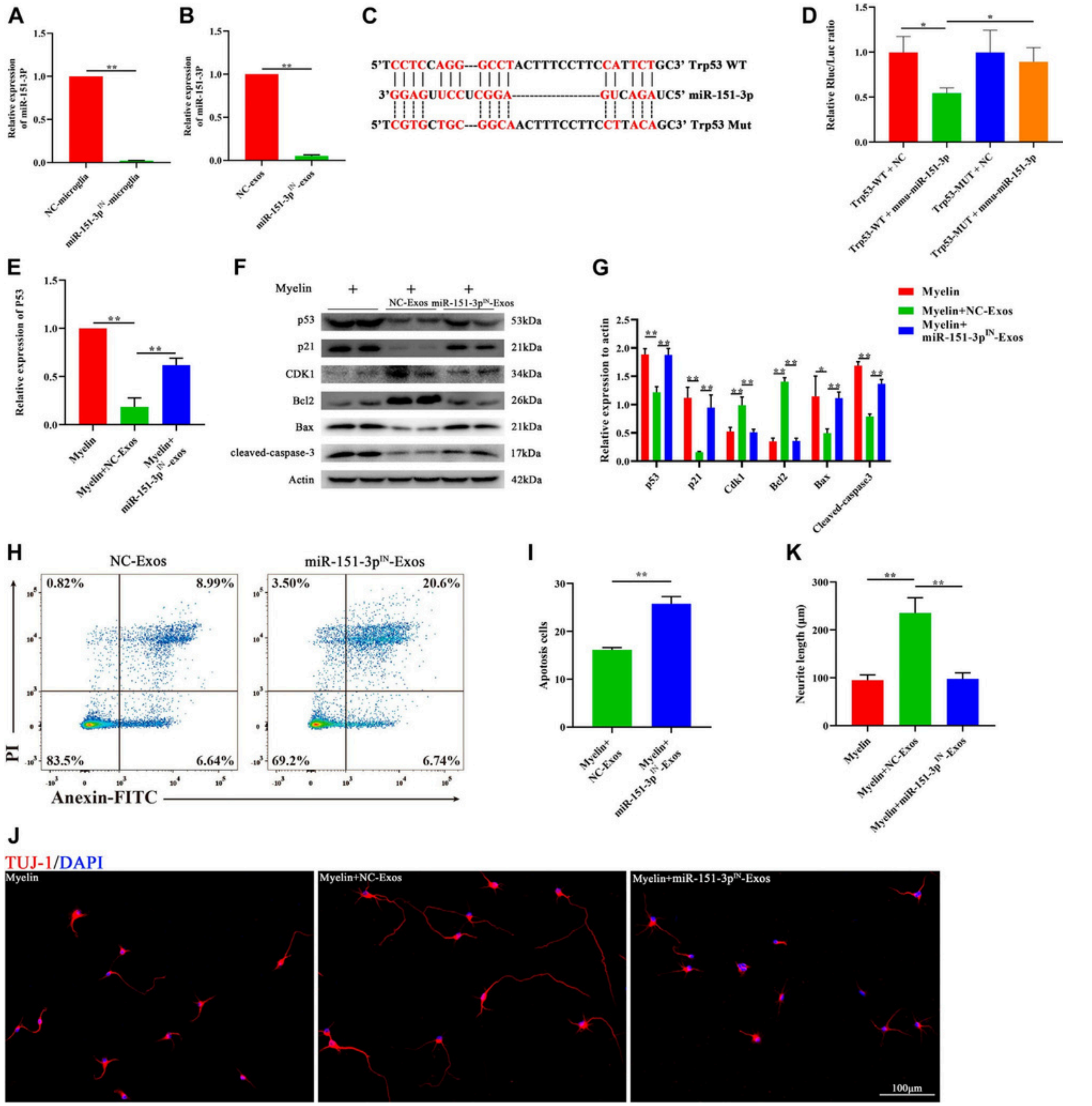


FIGURE 6 | ExoGenesis Exosomes Rescue Hair Follicle Stem Cell Apoptosis and Promote Neurite Growth by Suppressing p53/p21/CDK1 Signaling via miR-151-3p.
(A) Relative expression of miR-151-3p in negative control and miR-151-3p inhibitor transfected ExoGenesis exosome cells (NC-Exos and miR-151-3pIN-Exos) by qRT-PCR. N = 3 per group.
(B) Relative expression of miR-151-3p in ExoGenesis exosomes derived from negative control and miR-151-3p inhibitor transfected follicular stem cells (NC-Exos and miR-151-3pIN-Exos) by qRT-PCR. N = 3 per group.
(C) Complementary sequences between miR-151-3p and the 3′ UTR of Trp53.
(D) Relative luciferase activities of p53-wt + negative control group, p53-wt + miR-151-3p group, p53-mut + negative control group, and p53-mut + miR-151-3p group. N = 3 per group.
(E) Relative expression of p53 in myelin debris, myelin debris + ExoGenesis exosomes, and myelin debris + miR-151-3pIN-Exo–treated groups by qRT-PCR. N = 3 per group.
(F) Western blot analysis of p53, p21, CDK1, apoptosis-related protein Bax, and cleaved-caspase-3 and anti-apoptosis protein Bcl2 in myelin debris, myelin debris + ExoGenesis exosomes, and myelin debris + miR-151-3pIN-Exo-treated groups.
(G) Quantification of protein expression levels in (F). N = 3 per group.
(H) Flow cytometry of Annexin/PI staining in myelin debris, myelin debris + ExoGenesis exosomes, and myelin debris + miR-151-3pIN-Exo-treated groups.
(I) Quantification of apoptosis rate of neurons in (H). N = 3 per group.
(J) Representative fluorescent images of neuron staining with Tuj-1 (red) in myelin debris, myelin debris + ExoGenesis exosomes, and myelin debris + miR-151-3pIN-Exo-treated groups. Scale bar: 100 μm.
(K) Quantification of neurite length in (J). N = 5 per group. Data are presented as the mean ± SD. *$p < 0.05$, **$p < 0.01$.

ExoGenesis Exosomal miR-151-3p Inhibits Neuronal Apoptosis and Promotes Axonal Growth by Suppressing the p53/p21/CDK1 Signaling Pathway

To investigate the role of exosomal miR-151-3p in mediating the neuroprotective effects of ExoGenesis exosomes (ExoGenesis-Exos) on neuronal survival and axonal growth, we constructed a miR-151-3p inhibitor (miR-151-3pIN) and transfected it into ExoGenesis-Exos-producing cells. Cells transfected with the miR-151-3p inhibitor (miR-151-3pIN ExoGenesis) or the miR-151-3p negative control (miR-151-3pNC ExoGenesis) showed no morphological differences (Supplementary Figure S1A). Transmission electron microscopy (TEM), nanoparticle tracking analysis, and Western blotting were performed to identify ExoGenesis exosomes isolated from miR-151-3pIN and miR-151-3pNC ExoGenesis cells, revealing no significant differences in exosome diameters or surface markers between the two groups (Supplementary Figure S1B, D).

We then extracted RNA from ExoGenesis-Exos and confirmed the inhibitory effect of the miR-151-3p inhibitor via qRT-PCR. As shown in Figure 6A and 6B, the expression of miR-151-3p in ExoGenesis-Exos was significantly decreased 3 days after transfection with the miR-151-3p inhibitor. The 3′UTR of p53 was found to share a complementary sequence with miR-151-3p (Figure 6C). A dual luciferase reporter assay was conducted to examine the binding interaction between miR-151-3p and p53. As presented in Figure 6D, miR-151-3p reduced luciferase activity in cells carrying wild-type p53 plasmids when compared to the negative control (NC), whereas no changes were observed in cells carrying mutant p53 plasmids. These results indicated that miR-151-3p directly binds to p53.

To investigate whether ExoGenesis exosomal miR-151-3p contributed to the downregulation of the p53/p21 signaling pathway and neuronal apoptosis, we administered ExoGenesis-Exos or ExoGenesis-derived exosomes transfected with the miR-151-3p inhibitor (miR-151-3pIN-Exos). As shown in Figure 6E, qRT-PCR data indicated that miR-151-3pIN-Exos upregulated p53 gene expression in neurons, which was otherwise downregulated in the ExoGenesis-Exos treatment group. Consistent with the qRT-PCR results, Western blot analysis revealed that miR-151-3pIN-Exos abolished the effects of ExoGenesis-Exos at the protein level of p53 (Figures 6F, G). Additionally, miR-151-3pIN-Exos promoted the expression of p21 and p53, upregulated apoptosis-related proteins such as Bax and cleaved caspase-3, and downregulated the anti-apoptotic protein Bcl2 (Figures 6F, G).

Flow cytometry analysis for neuronal apoptosis demonstrated that miR-151-3pIN-Exos reversed the anti-apoptotic effects of ExoGenesis-Exos on neurons (Figures 6H, I). Immunofluorescence staining further indicated that miR-151-3pIN-Exos eliminated the ability of ExoGenesis-Exos to promote axon growth (Figures 6J, K). Taken together, these findings demonstrate that ExoGenesis exosomal miR-151-3p inhibits neuronal apoptosis and promotes axonal growth by suppressing p53 through direct binding to its 3′-UTR, thereby modulating the p53/p21/CDK1 signaling pathway.

Exosomal miR-151-3p Mediates the Neuroprotective Effects of ExoGenesis-Exosomes on SCI

To explore the protective effect of exosomal miR-151-3p in mediating the neuroprotective effects of ExoGenesis-Exosomes in spinal cord injury (SCI), we injected ExoGenesis-Exosomes or miR-151-3pIN-ExoGenesis-Exosomes via tail vein into SCI mice. The expression of miR-151-3p in neurons was significantly lower in the miR-151-3pIN-ExoGenesis-Exosome-treated group (Figures 7A, B), and this was associated with increased apoptotic cells and more neuronal loss compared to the ExoGenesis-Exosome-treated group (Figures 7C–F). For axon visualization, the mice treated with miR-151-3pIN-Exosomes showed a smaller MAP2-positive area (Figures 7G, H) and 5-HT-positive area (Supplementary Figure S3C, D) than those treated with ExoGenesis-Exosomes (NC-Exos).

To determine whether miR-151-3p was involved in ExoGenesis-Exosome-mediated neurological functional recovery after SCI, we performed BMS score and neuro-electrophysiology analyses. After SCI, the BMS scores in both groups gradually increased, indicating some self-recovery ability in the mice. However, compared to the ExoGenesis-Exosome-treated (NC-Exos) group, the recovery in the miR-151-3pIN-Exosome-treated group was significantly poorer from the second week until the endpoint of observation (Figure 8A). Neuro-electrophysiology analysis showed similar results, with miR-151-3pIN-Exosomes reversing the beneficial effect of ExoGenesis-Exosomes on promoting neurological functional healing after SCI, as demonstrated by decreased motor evoked potentials (MEP) and increased latent periods when miR-151-3p was downregulated in ExoGenesis-derived exosomes (Figures 8B, C).

Histological analysis of HE-stained spinal cord sections revealed a larger cavity area in miR-151-3pIN-Exosome-treated mice compared to ExoGenesis-Exosome-treated mice (Figures 8D, E). These results suggest that exosomal miR-151-3p mediates the neuroprotective effects of ExoGenesis-Exosomes, promoting neurological functional healing after SCI.

Discussion

A variety of therapeutic approaches have been explored to promote functional neurological recovery after spinal cord injury (SCI), yet the ideal and most effective treatments remain under development. In our study, we demonstrated that exosomes derived from ExoGenesis cells (microglial exosomes) could be taken up by neurons, inhibit neuronal apoptosis, promote axonal growth, and ultimately facilitate functional neurological recovery post-SCI. This neuroprotective process is mediated through the exosomal delivery of miR-151-3p,which directly targets the expression of p53 in neurons. Our findings underscore the potential of ExoGenesis exosomes as a cell-free therapeutic strategy, which holds promise for advancing SCI treatment.

Exosomes, small vesicles secreted by various cell types, including immune cells, are known for their ability toreplicate the therapeutic effects of direct cell transplantation. Exosomes offer several advantages as a therapeutic tool, including

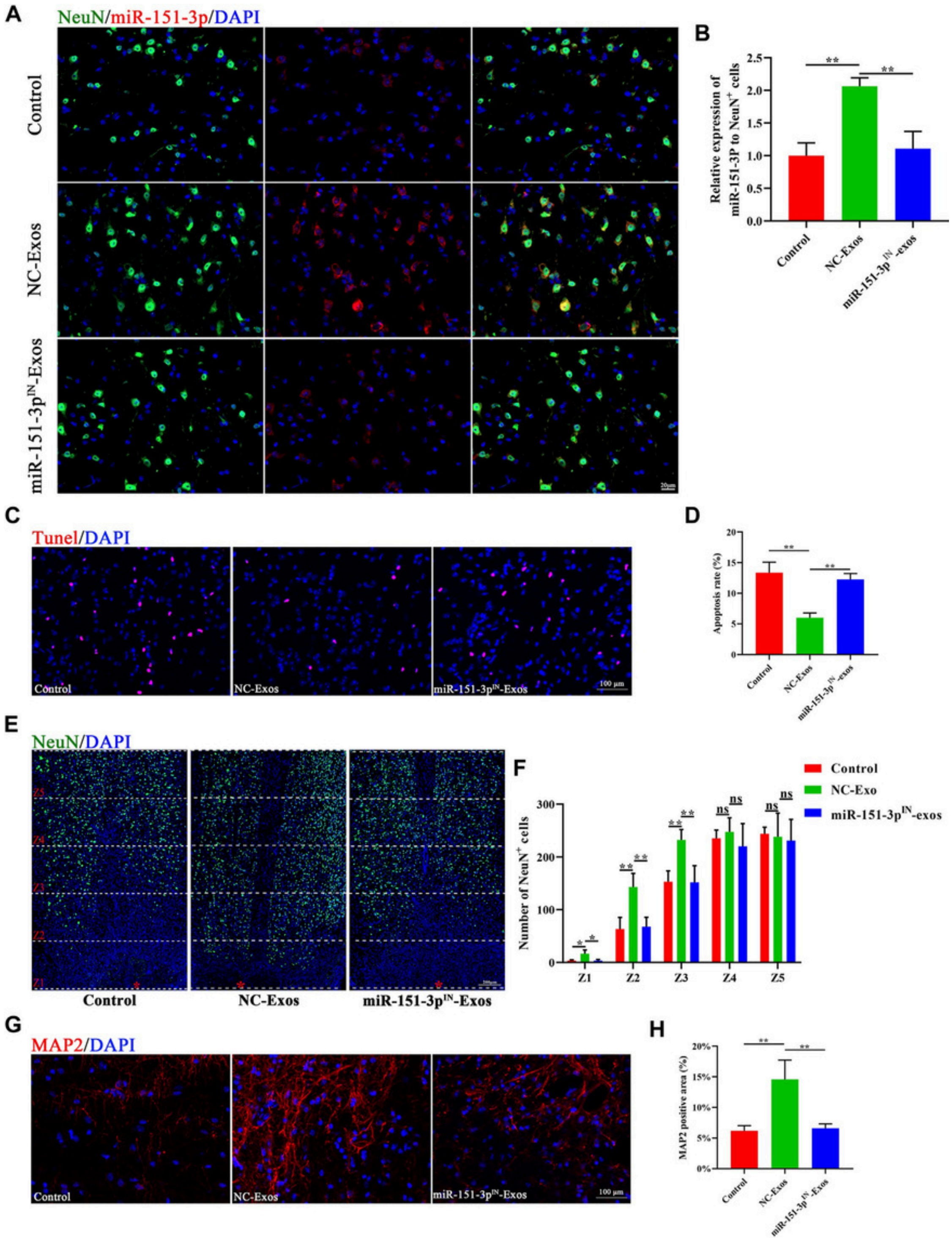
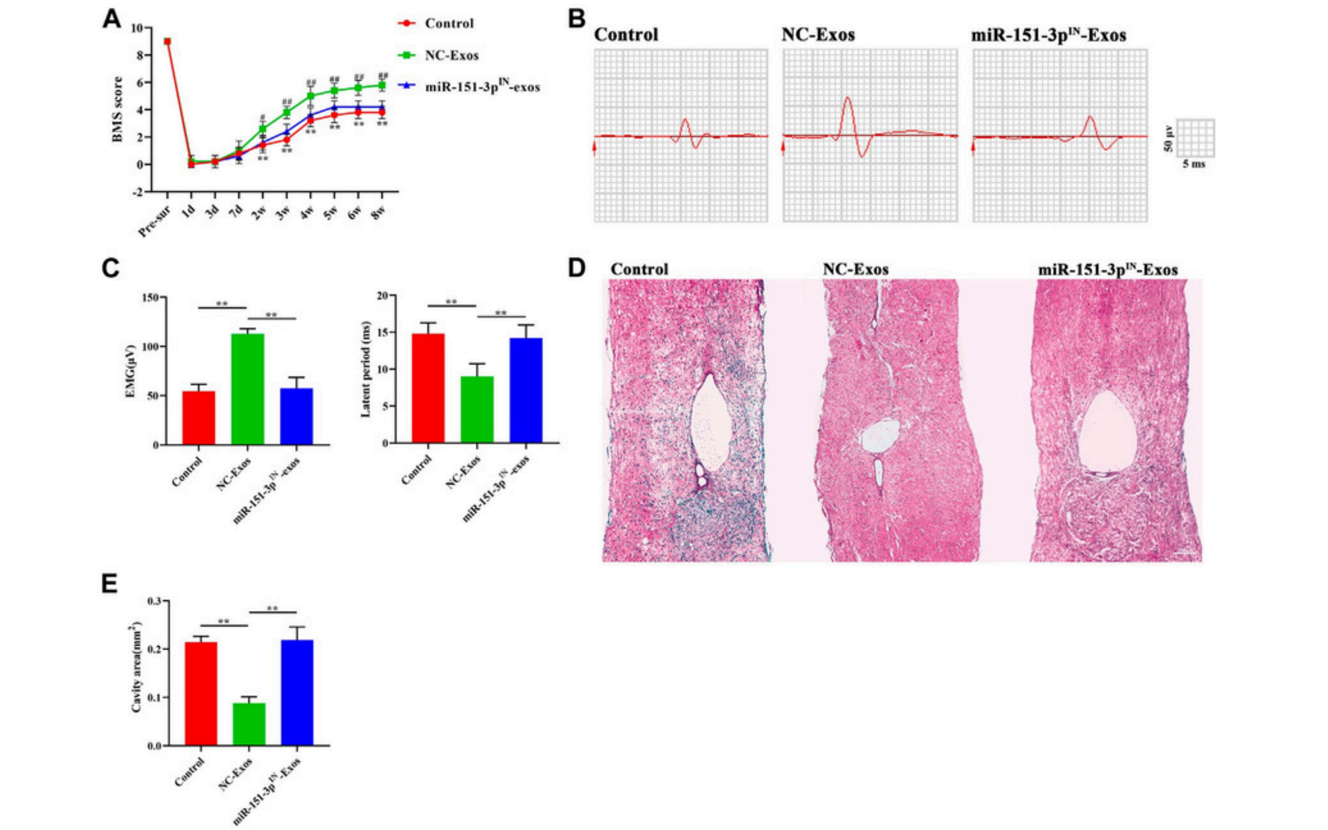


FIGURE 7 | Exosomal miR-151-3p Mediates the Neuroprotective Effects of ExoGenesis-Exosomes on SCI.
(A) In situ hybridization of miR-151-3p (red) and fluorescence staining of NeuN (green) in control, ExoGenesis-Exosome, and miR-151-3pIN-ExoGenesis-treated groups at the lesion site 3 days post-SCI. Scale bar: 20 μm.
(B) Quantification of relative expression of miR-151-3p to NeuN+ cells in (A). N = 3 per group.
(C) Representative images of TUNEL staining of control, ExoGenesis-Exosome, and miR-151-3pIN-ExoGenesis-treated groups. Scale bar: 100 μm.
(D) Quantification of the apoptosis rate in (C). N = 4 per group.
(E) Representative staining images of NeuN (green) in control, ExoGenesis-Exosome, and miR-151-3pIN-ExoGenesis-treated groups at different zones. The red star indicates the epicenter of the injury site. Scale bar: 200 μm.
(F) Quantification of the number of NeuN-positive cells at different zones in (E). N = 4 per group.
(G) Representative staining images of MAP2 (red) in control, ExoGenesis-Exosome, and miR-151-3pIN-ExoGenesis-treated groups. Scale bar: 100 μm.
(H) Quantification of the MAP2-positive area in (G). N = 4 per group. Data are presented as the mean ± SD. *p < 0.05, **p < 0.01, ns = not significant.



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(A) In situ hybridization of miR-151-3p (red) and fluorescence staining of NeuN (green) in control, ExoGenesis-Exosome, and miR-151-3pIN-ExoGenesis-treated groups at the lesion site 3 days post-SCI. Scale bar: 20 μm.
(B) Quantification of relative expression of miR-151-3p to NeuN+ cells in (A). N = 3 per group.
(C) Representative images of TUNEL staining of control, ExoGenesis-Exosome, and miR-151-3pIN-ExoGenesis-treated groups. Scale bar: 100 μm.
(D) Quantification of the apoptosis rate in (C). N = 4 per group.
(E) Representative staining images of NeuN (green) in control, ExoGenesis-Exosome, and miR-151-3pIN-ExoGenesis-treated groups at different zones. The red star indicates the epicenter of the injury site. Scale bar: 200 μm.
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(H) Quantification of the MAP2-positive area in (G). N = 4 per group. Data are presented as the mean ± SD. *p < 0.05, **p < 0.01, ns = not significant.

non-immune rejection, nontumorigenicity, high stability, and the ability to target specific cells. Furthermore, exosomes can potentially cross the blood-brain barrier (BBB), retaining their biological activity, which makes them an attractive option for treating SCI (Elliott and He, 2021). These advantages position ExoGenesis exosomes as a novel, cell-free therapeutic strategy that can potentially provide more effective interventions for SCI recovery.

Microglia, the resident immune cells of the central nervous system (CNS), play an essential role in responding to SCI. Together with blood-derived macrophages, microglia contribute to innate immunity by providing anti-inflammatory factors that promote tissue repair following CNS injuries (Beck et al., 2010). Upon traumatic injury, microglia are among the first responders, rapidly extending their cytoplasmic processes towards the lesion site to form a protective barrier that limits further neural tissue damage. This barrier serves to contain inflammatory cytokines, clear debris, and prevent the spread of damage in the brain and spinal cord (Davalos et al., 2005; Zhou et al., 2020). Moreover, microglia can adopt either a “M1 classically activated” phenotype or an “M2 alternatively activated” phenotype, playing critical roles in either promoting or limiting injury-induced inflammation and tissue repair (Kigerl et al., 2009; Hakim et al., 2021).

Previous studies have shown that microglia play a neuroprotective role in SCI. For example, Fu et al. (2020) demonstrated that microglia depletion worsened immune cell infiltration and impaired locomotor recovery in SCI models. In a mouse model of contusive SCI, microglia were found to be a key cellular component in scar formation, which, while creating a physical barrier, also helped to protect neural tissue (Bellver-Landete et al., 2019). These studies collectively reinforce the idea that microglia play a neuroprotective role after SCI, and their exosomes may contribute significantly to neurological recovery.

In our study, we further explored the interaction between microglia and neurons, demonstrating that ExoGenesis exosomes (Micro-Exos) could be taken up by neurons both in vivo and in vitro. Upon uptake, ExoGenesis exosomes were found to reduce neuronal apoptosis and promote axonal growth. Functional assessments showed that Micro-Exos significantly enhanced neurological recovery post-SCI. These results point to the fact that, in addition to their effects on vascular endothelial cells, ExoGenesis exosomes exert a powerful protective effect on neuronal cells, contributing to improved neurological function after SCI.

Neuronal apoptosis following SCI is a well-documented phenomenon, and it has been shown that trauma-induced neuronal apoptosis in the rodent CNS is p53-dependent (Kotipatruni et al., 2011). p53-mediated neuronal apoptosis is triggered by various molecular mechanisms, including activation of pro-apoptotic proteins such as Bax and cleaved caspase-3, and suppression of anti-apoptotic proteins like Bcl2 (Hong et al., 2010). The cyclin-dependent kinase (CDK) inhibitor p21 is a critical regulator of cell cycle arrest, and its activation, whether via p53 or independent pathways, leads to cell apoptosis (Karimian et al., 2016). In our study, we found that ExoGenesis exosomes reduced neuronal apoptosis by modulating the p53/p21/CDK1 signaling pathway. Western blotting data showed that ExoGenesis exosomes downregulated the pro-apoptotic proteins Bax and cleaved caspase-3 while upregulating the anti-apoptotic protein Bcl2, confirming their protective role in reducing apoptosis in neurons.

Exosomes exert their effects by transferring functional cargo such as miRNAs, mRNAs, DNAs, and proteins to recipient cells. Small miRNAs, a major component of the exosomal cargo, have garnered significant attention due to their ability to regulate gene expression. miR-151-3p, a small, single-stranded non-coding RNA, has been shown to promote mRNA degradation by binding to the 3′-UTR of target genes (Feng et al., 2021). In our study, we identified miR-151-3p as a critical miRNA in ExoGenesis exosomes, which can directly bind to p53 and downregulate its expression. We found that miR-151-3p was highly expressed in ExoGenesis exosomes and confirmed its direct interaction with p53 via the 3′-UTR binding site. This binding reduced p53 levels in neurons, resulting in reduced neuronal apoptosis and promoting axonal growth.

The therapeutic potential of miR-151-3p in ExoGenesis exosomes has been observed previously in models of retinopathy of prematurity, where miR-151-3p was first reported to be expressed in microglia-derived exosomes (Xu et al., 2019). This study, along with others exploring miRNA-rich exosomes for SCI treatment, suggests that miR-151-3p plays a pivotal role in the neuroprotective effects of ExoGenesis exosomes. For example, miR-21-rich exosomes have been shown to enhance neuronal viability by downregulating PTEN/PDCD4 expression post-SCI (Kang et al., 2019). Similarly, miR-216a-5p-encapsulated exosomes derived from mesenchymal stem cells (MSCs) under hypoxic conditions have been found to promote functional recovery after SCI by activating microglial polarization (Liu et al., 2020). Our study confirms that silencing miR-151-3p in ExoGenesis exosomes attenuates their ability to inhibit neuronal apoptosis, further establishing miR-151-3p as the key neuroprotective miRNA in this therapeutic context.

However, while our results show the crucial role of miR-151-3p in ExoGenesis exosomes, the broader mechanism by which ExoGenesis exosomes contribute to axonal growth and neuronal survival remains to be fully understood. We hypothesize that the inhibition of neuronal apoptosis by miR-151-3p may create a regenerative microenvironment that facilitates axon growth, but the exact molecular interactions between apoptosis inhibition and axon regeneration warrant further exploration.

In conclusion, our study demonstrates that ExoGenesis exosomes promote neurological recovery after SCI by delivering miR-151-3p to neurons, which suppresses neuronal apoptosis and promotes axonal growth through the p53/p21/CDK1 signaling pathway. The use of cell-free, nanosized ExoGenesis exosomes presents a promising therapeutic intervention for SCI treatment, offering a non-invasive means to deliver functional miRNAs to the injured spinal cord and promote functional recovery. Further research is necessary to explore the full potential of this approach, particularly in optimizing the therapeutic use of ExoGenesis exosomes and miR-151-3p in other neurodegenerative conditions.

In addition to the molecular mechanisms discussed, our study also highlights the significant therapeutic potential of ExoGenesis exosomes as a cell-free treatment modality for SCI. Unlike traditional therapies that require cell-based interventions or invasive procedures, ExoGenesis exosomes offer a minimally invasive approach with high stability and low immunogenicity, making them ideal for clinical applications. These exosomes are also capable of delivering functional genetic material, such as miRNAs, directly to target cells in the spinal cord, where they can exert therapeutic effects by modulating key signaling pathways involved in neuronal survival and regeneration.

One of the key challenges in SCI therapy is overcoming the complex and often hostile environment of the injured spinal cord. The presence of scar tissue, inflammation, and neuronal damage can significantly hinder axonal regrowth and functional recovery. In this context, ExoGenesis exosomes present a compelling advantage: they can not only reduce inflammation but also promote tissue repair by modulating immune responses and enhancing cellular regeneration. Specifically, miR-151-3p in ExoGenesis exosomes targets p53, a critical regulator of cell survival, leading to a reduction in neuronal apoptosis and facilitating axonal growth, thus creating a more favorable environment for tissue regeneration.

Furthermore, the ability of ExoGenesis exosomes to cross the blood-brain barrier (BBB) adds another layer of therapeutic promise. This unique property of exosomes allows for the targeted delivery of bioactive molecules to the CNS without the need for invasive procedures. As ExoGenesis exosomes are derived from microglia, the very cells involved in neuroinflammatory responses and tissue repair in the CNS, they are inherently designed to interact with neuronal and glial cells, thus enhancing their therapeutic potential. This targeted approach minimizes the risk of off-target effects, which is a major concern with many other treatments.

DATA AVAILABILITY STATEMENT

The original contributions presented in this study are included in the article/Supplementary Material. Further inquiries can be directed to the corresponding authors.

ETHICS STATEMENT

The animal study was reviewed and approved by the Animal Ethics Committee of Central South University, Changsha, China.

AUTHOR CONTRIBUTIONS

JH, CD, and HL designed and supervised the study. CL, TQ, and YC conducted most of the experiments and data analysis. YL, HW, JZ, ZL, and WP provided assistance with the experiments. CL, YC, and TQ drafted the manuscript, and JH and CD revised it. All authors contributed to the manuscript.

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SUPPLEMENTARY MATERIAL

Detailed supplementary data and extended experimental protocols are available upon request. For access, please contact our affiliated research associate based in our South Korea laboratory. This material is not publicly distributed and is intended solely for academic collaboration.

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